

Question	Answer	Mark	Mark scheme	Additional guidance
21 (a)	$\frac{4-3x}{x}$	M1	first step to change the subject of $y = \frac{4}{x+3}$ or $x = \frac{4}{y+3}$ eg $y(x+3) = 4$ or $x(y+3) = 4$	May be seen as part of an equation
		A1	$\frac{4-3x}{x}$ for $\frac{4}{x}$ oe eg $\frac{4}{x} - 3$	
		P1	for method to find gf(x), eg $3 \times \frac{4}{x+3} + 1$ oe or gf(2a) eg $3 \times \frac{4}{2a+3} + 1$ oe	
		P1	for setting up equation and reducing to form $ax^2 + bx + c (= 0)$, eg $2a^2 + a - 15 (= 0)$ or $-2a^2 - a + 15 (= 0)$	
		P1	(dep on P2) for process to solve quadratic equation eg $(2a-5)(a+3) (= 0)$ $\frac{-1 \pm \sqrt{1^2 - 4 \times 2 \times -15}}{2 \times 2}$	
		A1	or for critical values $-3, 2.5$	
(b)	$a > 2.5$	A1	for any statement that a is greater than 2.5	Need not be given as an inequality statement